

SRI VIGNESWARAN K

AI Engineer · LLM & GenAI Specialist · MLOps

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PROFESSIONAL SUMMARY

Results-driven AI Engineer with hands-on experience building production-grade LLM pipelines, AI Agents, and Retrieval-Augmented Generation (RAG) systems. Proficient in Python, LangChain, OpenAI APIs, and vector databases, with real-world experience deploying GenAI solutions that cut manual effort by 20%+. Strong foundation in NLP, data engineering, and SAP GenAI platforms. Passionate about designing scalable, explainable AI systems that solve concrete business problems.

WORK EXPERIENCE

AI Engineer Intern (GenAI & LLM). [Sierra Digital Inc.](#)

Coimbatore, India | Jun 2024 – Dec 2024

- Architected and deployed AI Agent workflows using LangChain and OpenAI GPT models to automate extraction of structured insights from unstructured enterprise data — reducing analyst query time by ~30%.
- Built a production RAG pipeline using SAP GenAI and vector-based retrieval to answer natural-language queries against 10K+ rows of client datasets with high factual accuracy.
- Designed and fine-tuned prompt engineering strategies (chain-of-thought, few-shot, role prompting) to improve LLM output quality and reduce hallucinations across multiple use cases.
- Implemented an OCR + LLM document intelligence pipeline to process 50+ unstructured documents (PDFs, scanned files), improving data availability and cutting manual reporting time by 20%.
- Developed structured metadata and data context layers to improve LLM extraction accuracy and enable consistent, schema-aware outputs across diverse datasets.
- Built 3+ KPI dashboards in SAP Analytics Cloud (SAC) to visualize AI pipeline output, performance trends, and data quality metrics for business stakeholders.
- Conducted data quality validation and preprocessing (cleaning 10K+ records) to ensure reliable inputs for downstream AI models and reporting systems.

AI / ML PROJECTS

AI-Powered Abusive Comment Detector | *Python, Transformers (BERT), NLP, Scikit-learn*

- Fine-tuned a BERT-based classifier on a custom labeled dataset to detect abusive/offensive language with 91%+ F1 score.
- Built preprocessing pipeline (tokenization, stopword removal, augmentation) and deployed model via a Flask inference API.
- Implemented explainability using LIME to surface token-level attributions, making model decisions interpretable for moderation teams.

Autonomous AI Agent with Tool Use | *Python, LangChain Agents, OpenAI Functions, Serper API*

- Developed a ReAct-pattern AI agent capable of reasoning over multi-step tasks using tool calls (web search, calculator, code execution).
- Integrated OpenAI function-calling to enable structured, reliable tool dispatch and response parsing with zero prompt injection risk.

Customer Insight NLP Dashboard | *Python, spaCy, Pandas, Power BI, REST API*

- Built an NLP pipeline (entity extraction, sentiment analysis, topic modeling with LDA) to process customer feedback at scale.
- Automated report generation and connected NLP output to an interactive Power BI dashboard exposing actionable business insights.

TECHNICAL SKILLS

AI / LLM: LangChain, OpenAI API (GPT-4, Embeddings, Functions), Prompt Engineering, RAG, AI Agents, Fine-tuning
ML / NLP: Scikit-learn, BERT, Text Classification, OCR
Vector & Data: Pandas, NumPy, SQL, Data Cleaning & Validation, Metadata Management
Platforms & Tools: SAP GenAI, SAP Analytics Cloud, SAP S/4HANA, SAP Datasphere, Power BI,
APIs & Backend: Flask, REST APIs, JSON, Python scripting
DevOps / Infra: Git, GitHub, VS Code, Jupyter Notebooks
Languages: Python, SQL, basic JavaScript

EDUCATION

M.Tech – Computer Science Sri Ramakrishna Engineering College CGPA: 7.08 / 10.0	<i>May 2026</i>
Higher Secondary – Computer Science Sri Ramakrishna Matriculation Higher Secondary School Grade: 81.2%	<i>May 2021</i>

CERTIFICATIONS

- ▶ NLP Specialization – Coursera (deeplearning.ai)
- ▶ Data Analytics Virtual Experience – Deloitte, Forage
- ▶ Getting Started with Power BI – Microsoft
- ▶ Introduction to End-to-End Data Analysis – Microsoft
- ▶ Discover Data Analysis – Microsoft